

Progress Report: Smart-ASHA Research Paper Development

Project Title: Smart-ASHA: A Multi-Modal, Edge-AI Cardiopulmonary & Biochemical Triage Framework for Rural Healthcare

Target Standard: IEEE Conference/Journal (6–8 Pages, Double-Column)

Date: October 2023 – Current

Status: In-Progress (Drafting Phase)

1. Executive Summary

The Smart-ASHA project aims to bridge the rural healthcare gap by providing ASHA workers with a low-cost, offline, Edge-AI-powered triage tool. To date, we have completed the foundational research, established the paper's structural architecture, and generated an initial 4-page technical draft. The project is now moving into the "Technical Density" phase, focusing on mathematical modeling and experimental validation.

2. Completed Milestones

✓ Phase 1: Literature Synthesis & Domain Mapping

We conducted an exhaustive review of 20–30 peer-reviewed papers across four critical domains:

- **Edge AI & TinyML:** Analyzed resource-constrained inference for medical diagnostics.
- **Cardiopulmonary AI:** Studied CNN-based lung sound classification and ECG arrhythmia detection.
- **Non-Invasive Biochemical Sensing:** Researched spectral analysis for hemoglobin and metabolic detection.
- **Multimodal Fusion:** Evaluated frameworks for combining disparate physiological signals on embedded hardware.

✓ Phase 2: Structural Architecture & IEEE Compliance

We finalized a 15-point structural framework designed to meet the rigorous requirements of IEEE publications:

1. Title & Abstract | 2. Keywords | 3. Introduction | 4. Literature Review | 5. Problem Statement | 6. Proposed Architecture | 7. Methodology | 8. Implementation | 9. Results | 10. Discussion | 11. Applications | 12. Future Scope | 13. Conclusion | 14. References | 15. Appendices.

✔ **Phase 3: Initial Drafting & Prototyping**

- **Draft Generation:** Produced a 3–4 page IEEE double-column template.
- **Visual Strategy:** Identified key figure placements (Architecture Block Diagram, System Overview, and Performance Graphs).
- **Artifact Output:** Generated a preliminary PDF (Smart_ASHA_Draft_Research_Paper.pdf) to serve as the baseline for expansion.

3. Current Technical Status

Metric	Status	Details
Page Count	~4 Pages	Baseline content established; expansion required.
Literature Review	80% Complete	Context is strong; needs a comparative table.
Mathematical Rigor	20% Complete	Placeholders for STFT and CNN logic exist.
Experimental Data	10% Complete	Numerical results currently use simulated/mock values.
References	50% Complete	Sources identified; IEEE citation formatting pending.

4. Technical Highlights of the Framework

- **Modality:** Multi-modal (ECG + Lung Audio + PPG-based Biochemical).
- **Hardware Target:** Edge-AI (TinyML) on microcontrollers (e.g., ESP32/ARM Cortex-M).
- **Novelty:** Addressing the specific "Offline/Low-Power" requirement for ASHA workers in remote environments.
- **Signal Processing:** Utilization of Short-Time Fourier Transform (STFT) for audio and Wavelet Transform for ECG.

5. Gap Analysis (The Road to 8 Pages)

To achieve publication readiness, the following "Value Additions" are required:

1. **Mathematical Formulations:** Formalizing the sensor fusion algorithm using weighted decision matrices.
2. **Model Optimization Details:** Documenting the quantization (Float32 to INT8) and pruning

processes used to fit the model on-device.

3. **Experimental Validation:** Developing comprehensive tables for Accuracy, Precision, Recall, and F1-Score across different triage categories.
4. **Hardware Evaluation:** Documenting latency (ms) and power consumption (mAh) benchmarks.

6. Upcoming Action Plan

- **Short Term:** Integrate mathematical modeling for signal preprocessing and TinyML architecture.
- **Mid Term:** Expand the "Results" section with simulated or experimental performance metrics and confusion matrices.
- **Long Term:** Finalize IEEE citation formatting and perform a plagiarism/quality check for submission.

Report generated for the Smart-ASHA Development Team.